



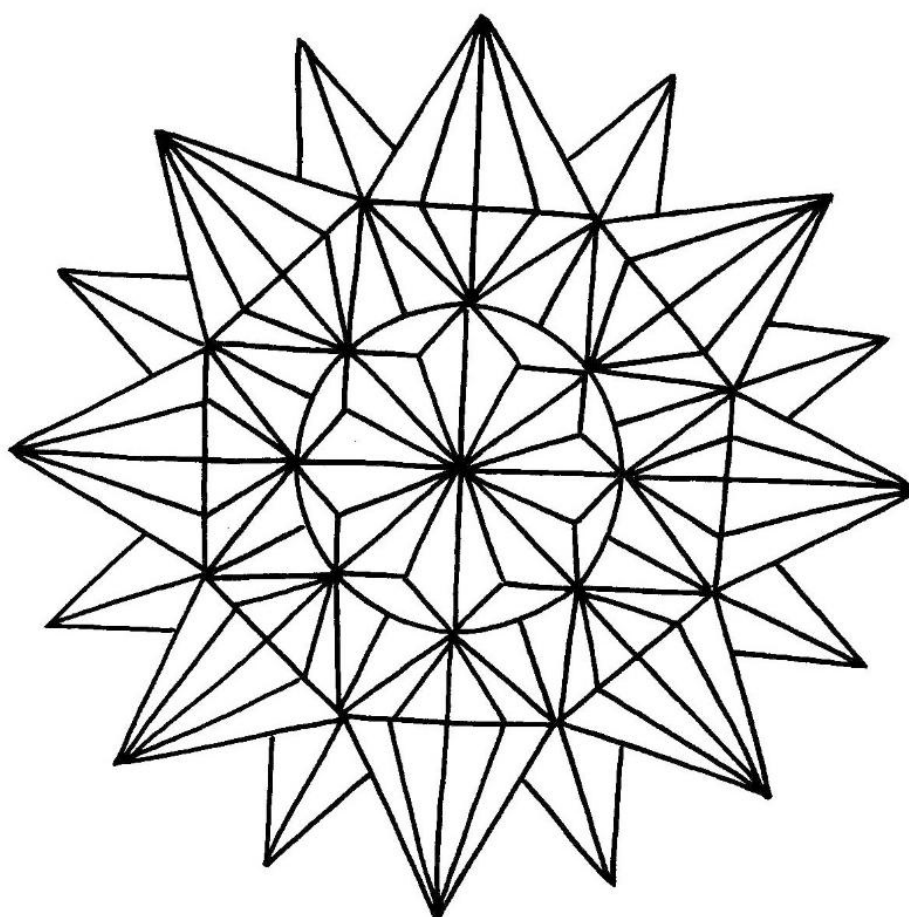
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Tour de Maths 2018

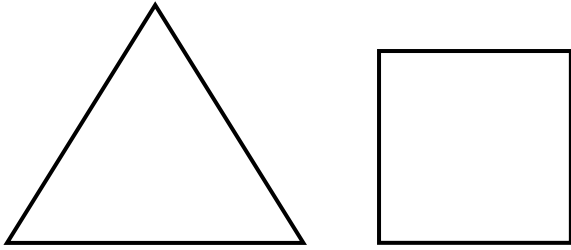
Leg 1

Question Paper- MEMO



5 Mark Questions

Question 1



An equilateral triangle and a square have equal areas.

Which statement is the most accurate about their perimeters?

- a) The triangle has the larger perimeter
- b) The square has the larger perimeter
- c) The perimeters are equal
- d) Not enough information has been provided

MEMO

Given information

Let: $side\Box = x$
 $side\Delta = y$

Use Pythag to find height of triangle:

$$\begin{aligned} h &= \sqrt{y^2 - \frac{1}{4}y^2} \\ &= \sqrt{\frac{3}{4}y^2} \\ &= \frac{\sqrt{3}}{2}y \end{aligned}$$

Perimeters:

$$Perimeter\Delta = 3y$$

$$Perimeter\Box = 4x$$

Areas:

$$Area\Delta = \frac{1}{2} \times \text{base} \times \text{height}$$

$$Area\Box = \text{length} \times \text{length}$$

$$\frac{1}{2} \times \text{base} \times \text{height} = \text{length}^2$$

$$\frac{1}{2} \times y \times \frac{\sqrt{3}}{2}y = x^2$$

$$\frac{\sqrt{3}y^2}{4} = x^2$$

$$\frac{\sqrt{3}}{2}y = x$$

$$2\sqrt{3}y = 4x$$

$$2,6321y = 4x$$

$$2,6321y \neq 3y$$

Therefore the square perimeter is smaller

ANSWER: A

Question 2

A farmer has a total of 242 sheep and goats.
 He sells half of his goats and buys 28 more sheep.
 He now has 3 times more sheep than goats.

How many sheep did he have originally?

MEMO

let sheep = s

goats = g

$s + g = 242$originally

$s = 242 - g$

$(s + 28) = 3\left(\frac{g}{2}\right)$after sales and purchase

sub

$$242 - g + 28 = \frac{3}{2}g$$

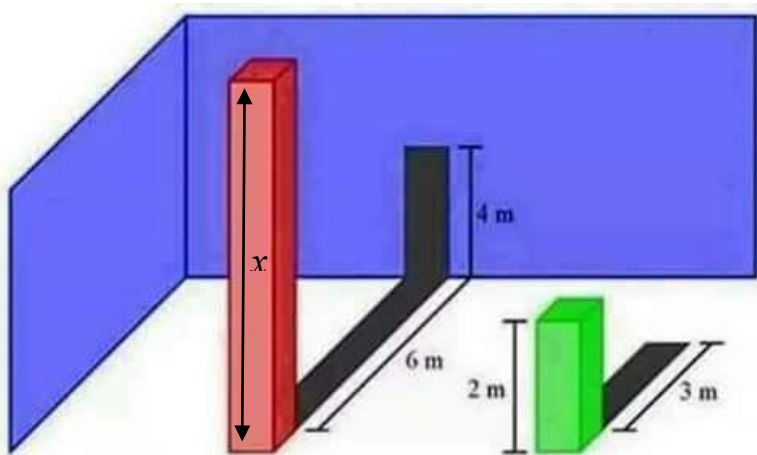
$$270 = \frac{5}{2}g$$

$$g = 108$$

$$\therefore s = 242 - 108$$

$$s = 134$$

ANSWER: 134

Question 3**MEMO**

Small tower gives shadow proportion of:

2:3

If we maintained this, then the shadow that is 4m high would cast its own shadow of:

$$4 \times \frac{3}{2} = 6m$$

so the long shadow would have been 12m

the tower therefore would have been:

$$12 \times \frac{2}{3} = 8m$$

ANSWER: 8m

Question 4

How can you add eight 8s to get the number 1000?

You may only use addition, and the digit 8 eight times.
Write out the whole sum on the answer sheet.

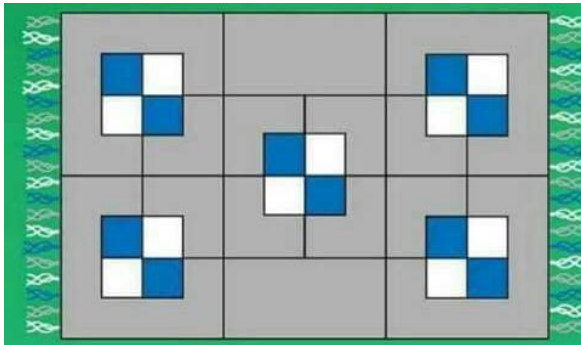
MEMO

$$888 + 88 + 8 + 8 + 8 = 1,000$$

ANSWER: AS ABOVE

Question 5

How many squares can you find in the picnic blanket?

**MEMO**

largest squares = 2
mid size = 9
4x4 size = 13
smallest = 20
total = 44

ANSWER: 44

Question 6

Can you see how the numbers in each row of blocks follow on from each other?

3	1	4	5
---	---	---	---

2	3	5	8
---	---	---	---

Use the same method to find the 2 missing numbers in the blocks below:

5			17
---	--	--	----

MEMO

The pattern seems to follow that

In 3145

$$5 + 3 = 8$$

$$8 \div 2 = 4 \text{ the 3rd cell}$$

$$4 - 3 = 1 \text{ the 2nd cell}$$

In 2358

$$2 + 8 = 10$$

$$10 \div 2 = 5 \text{ the 3rd cell}$$

$$5 - 2 = 3 \text{ the 2nd cell}$$

In 5 _ _ 17

$$5 + 17 = 22$$

$$22 \div 2 = 11 \text{ the 3rd cell}$$

$$11 - 5 = 6 \text{ the 2nd cell}$$

ANSWER:

5	6	11	17
---	---	----	----

Question 7

$$2 + 2 = 2 \times 2$$

$$1 + 2 + 3 = 1 \times 2 \times 3$$

In the above equations, there are respectively 2 and 3 whole numbers whose sum is equal to their product.

Find 4 whole numbers whose sum is equal to their product.

Write down only the sum (the product) on the answer sheet.

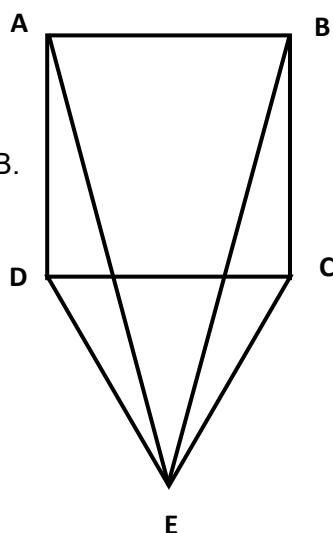
MEMO

$$1 + 1 + 2 + 4 = 1 \times 1 \times 2 \times 4$$

ANSWER: 8**Question 8**

ABCD is a square.
CDE is an equilateral triangle.

Determine the size of angle AEB.

**MEMO**

$$\widehat{ADE} = 90^\circ \text{ given}$$

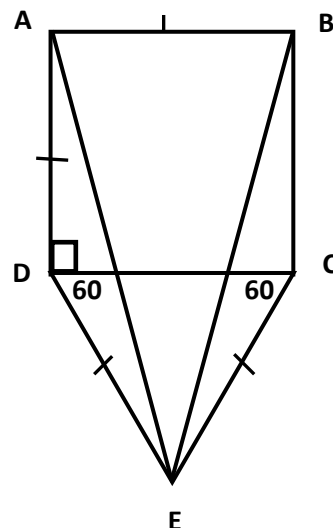
$$\widehat{CDE} = 60^\circ \text{ properties of equilateral triangle}$$

$$\widehat{AED} = 15^\circ \text{ properties of isos triangle}$$

$$\therefore \widehat{BEC} = 15^\circ$$

$$\widehat{AEB} = 60^\circ - 15^\circ - 15^\circ$$

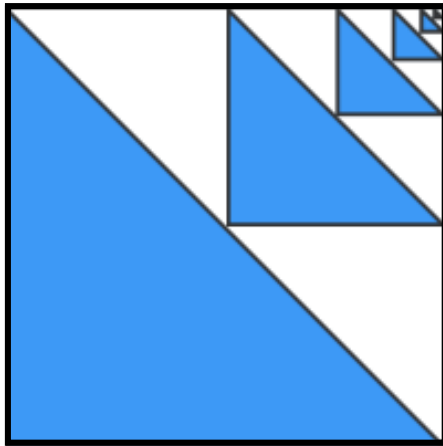
$$= 30^\circ$$

**ANSWER: 30**

Question 9

If the shaded triangles continue to be formed indefinitely, what fraction of the big square will be shaded?

- a) $\frac{2}{3}$
- b) $\frac{3}{4}$
- c) $\frac{4}{5}$
- d) $\frac{5}{6}$

**MEMO**

$$T_1 : \frac{1}{2} = 0,5$$

$$T_2 : \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 0,625$$

$$T_3 : \frac{1}{2} + \frac{1}{8} + \frac{1}{32} = \frac{21}{32} = 0,656$$

$$T_4 : \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} = \frac{85}{128} = 0,664$$

$$T_5 : \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} + \frac{1}{512} = \frac{341}{512} = 0,666$$

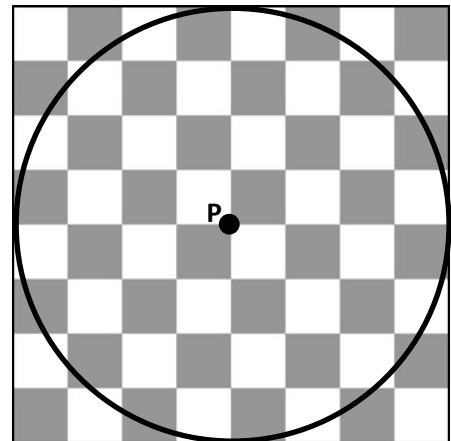
tending towards $\frac{2}{3}$

ANSWER: $\frac{2}{3}$

Question 10

A circle with a radius of 4 squares is drawn on the chess board with the centre at point P.

How many squares will have the circumference (edge) of the circle cut through them?



ANSWER: 28

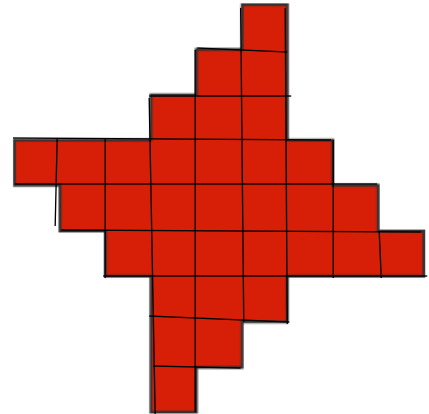
10 Mark Questions

Question 11

In the polygon, all adjacent edges are at right angles so that all interior angles are either 90° or 270° .
The four longer edges are equal in length and all of the shorter edges are also equal in length.
The area of the shape is 528 cm^2 .

What is the perimeter?

MEMO



$$528 \div 33 = 16 \text{ cm}^2 \text{ per unit}$$

$$16 = 4^2$$

therefore one edge of each square is 4cm

36 edges

$$36 \times 4 = 144 \text{ cm}$$

ANSWER: 144cm

Question 12

At a party everyone shook hands with everybody else.
There were 66 handshakes in total.

How many people were at the party

MEMO

$$\frac{n(n-1)}{2} = 66$$

$$n(n-1) = 132$$

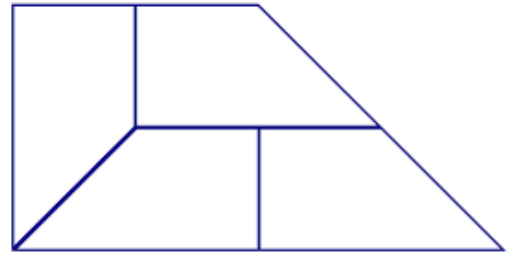
$$12(11) = 132$$

$$n = 12$$

ANSWER: 12

Question 13

Divide the trapezium into 4 equal, identical shapes by drawing in the necessary lines.

MEMO

ANSWER: AS ABOVE

Question 14

Two gears, one with 12 teeth and the other with 20 teeth, fit together as shown in the illustration. After how many rotations of the smaller gear will the two gears be back in exactly the same position?

MEMO

	Small Tooth	Large tooth
start	1	1
1	1	$1+12-(0 \times 20)=13$
2	1	$13+12-(1 \times 20)=5$
3	1	$5+12-(0 \times 20)=17$
4	1	$17+12-(1 \times 20)=9$
5	1	$9+12-(0 \times 20)=21=1$

ANSWER: 5 Rotations

Question 15

Twin primes are pairs of prime numbers which differ by two. The first twin primes are {3,5}, {5,7}, {11,13} and {17,19}.

Find the two pairs of twin primes that lie between 50 and 100.

MEMO

ANSWER: (59,61) and (71,73)

20 Mark Questions

Question 16

Using only 4 cuts, what is the maximum number of pieces that one can cut a pizza into?

MEMO



$$f(1) = 2$$

$$f(2) = 2 + f(1)$$

$$f(n) = n + f(n-1)$$

$$f(n) = n + (n-1) + \dots + 2 + f(1)$$

$$= f(1) + \sum_{k=2}^n k$$

$$= 2 + \frac{1}{2}(n+2)(n-1)$$

$$f(n) = \frac{1}{2}(n^2 + n + 2)$$

$$f(4) = \frac{1}{2}(4^2 + 4 + 2)$$

$$= 11$$

ANSWER: 11

Question 17

One day, Alice, Betty, Cathy, David and Edward were at home playing soccer. Unfortunately, one of them accidentally broke a window. When their mom asked them who did it they said the following:

Alice: "Betty broke it!"

Betty: "Alice is lying!"

Cathy: "I did it!"

David: "I didn't break it and Betty is lying."

Edward: "David is telling the truth."

Given that **at least three** of the children are telling the truth, who broke the window?

MEMO

Person	Statement	If it was....				
		Alice	Betty	Cathy	David	Edward
Alice	"Betty broke it"	F	T	F	F	F
Betty	"Alice is not telling the truth"	T	F	T	T	T
Cathy	"I did it"	F	F	T	F	F
David	"I did not break it and Betty is lying"	F	T	F	F	F
Edward	"David is telling the truth"	F	T	F	F	F

ANSWER: BETTY

Question 18

In the grid above, the number of gold coins in each block is given. Mahube needs to start in the bottom-left square (Start) and move to the top-right square (Finish).

She can only move up and to the right.

What is the maximum number of gold coins she can collect?

MEMO

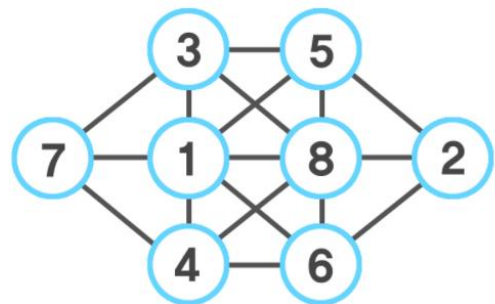
5	1	0	4	1	1	2	0
0	3	2	1	0	3	0	1
4	3	0	6	5	0	1	0
3	1	0	3	4	0	1	3
0	5	2	0	1	1	5	1
2	1	6	1	6	0	2	1
0	0	4	3	2	3	0	2

ANSWER: 32**Question 19**

I want to fill in the eight circles below using each of the numbers 1, 2, 3, 4, 5, 6, 7 and 8 **exactly once**.

Additionally, consecutive numbers – e.g. (1, 2) or (5, 6) – cannot be placed in circles which are connected by a line segment.

Given these constraints, what is the sum of the two numbers in the middle (i.e. $\alpha + \beta$)?

MEMO

The reasoning is:

If we choose a number other than 1 or 8 and put it into the center of the diagram (α, β), then, there are two numbers that are adjacent to it.

However, there is only one circle that is not connected to the number, and therefore, it is not possible to fill the diagram.

Then there're only 1 and 8 that can go into the two center circles, so we already know the answer is 9 given that such arrangement(s) of numbers exist(s).

ANSWER: 9

Question 19 The factorial symbol (!) indicates the result of multiplying a sequence of descending natural numbers down to 1. For example: $4! = 4 \times 3 \times 2 \times 1 = 24$ $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$ Simplify and give your answer as an integer:	MEMO $\frac{(10!+9!)(8!+7!)(6!+5!)(4!+3!)(2!+1!)}{(10!-9!)(8!-7!)(6!-5!)(4!-3!)(2!-1!)}$ $= \frac{9!(10+1).7!(8+1).5!(6+1).3!(4+1).(3)}{9!(10-1).7!(8-1).5!(6-1).3!(4-1).(1)}$ $= \frac{(11)(9)(7)(5)(3)}{(9)(7)(5)(3)(1)}$ $= 11$ ANSWER: 11
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Total: 200

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